Customer Behavior Analysis for Alfido Tech

Task 1 Objective

Analyze e-commerce customer transactions, identify purchasing patterns, segment customers using Recency, Frequency and Monetary (RFM) analysis, examine recorded churn patterns, and recommend actions to improve engagement and retention.

1. Introduction

This notebook-style report documents the customer behavior analysis with Python code snippets, summarized outputs, visual analysis, recommendations, and limitations. It follows the black-and-white layout used for Task 2.

2. Import Libraries and Load Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(
    "/kaggle/input/customer-behavior-analysis/"
    "ecommerce_customer_data_large.csv"
)
print("Dataset shape:", df.shape)
display(df.head())
df.info()
```

Observed output: 250,000 rows and 13 columns, including Customer ID, Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount, Payment Method, Customer Age, Returns, Customer Name, Age, Gender, and Churn.

3. Data Overview and Quality Checks

```
print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
print("Unique customers:", df["Customer ID"].nunique())
print("Total purchase amount:",
      df["Total Purchase Amount"].sum())
```

Observed findings: 49,661 unique customers; dates from 1 January 2020 to 13 September 2023; total recorded purchase amount 681,346,299 (currency not specified); no duplicate rows. Returns has 47,382 missing values (18.95%).

4. Data Cleaning and Preparation

```
df["Purchase Date"] = pd.to_datetime(
    df["Purchase Date"], errors="coerce"
)
# Keep missing Returns values visible; do not assume
# a missing value means zero.
df["Purchase Month"] = df["Purchase Date"].dt.to_period("M")
```

Returns values were not imputed because the meaning of missing values is not confirmed. The final month is partial because data ends on 13 September 2023.

5. Exploratory Data Analysis

```
display(df.describe(include="all"))

category_summary = (
    df.groupby("Product Category")["Total Purchase Amount"]
      .sum().sort_values(ascending=False)
)
monthly_summary = (
    df.groupby("Purchase Month")["Total Purchase Amount"].sum()
)
```

The analysis examined transaction summaries, purchase amounts by product category, and monthly purchasing trends. Exact category totals are not reproduced here because they are not present in the retained summary.

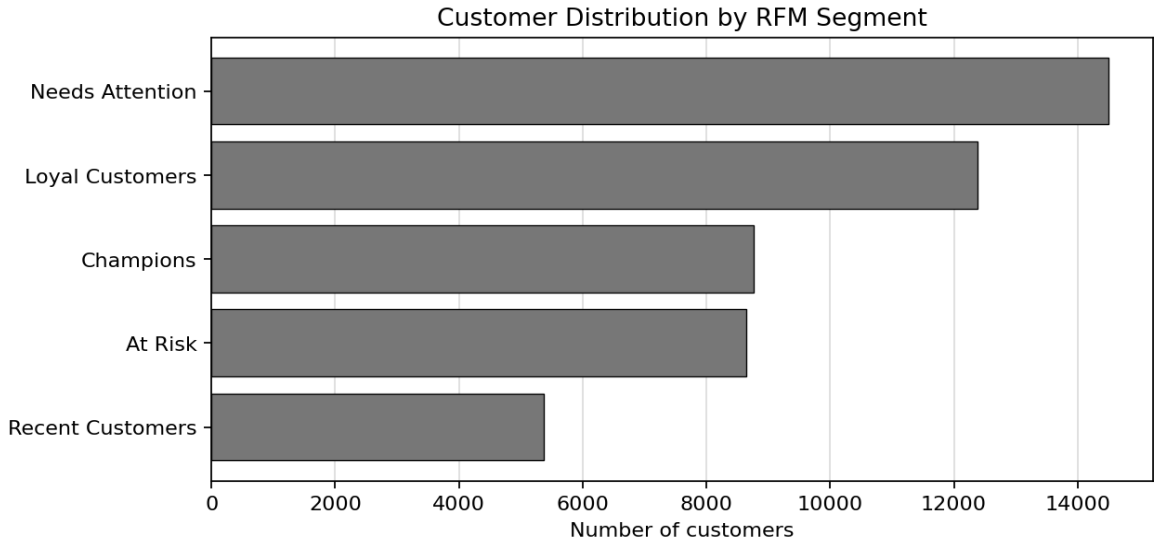
6. RFM Customer Segmentation

RFM evaluates Recency (how recently a customer purchased), Frequency (how often they purchased), and Monetary value (how much they spent). The segments below are rule-based groups, not a validated churn prediction model.

```
reference_date = df["Purchase Date"].max() + pd.Timedelta(days=1)
rfm = df.groupby("Customer ID").agg(
    Recency=("Purchase Date", lambda x:
             (reference_date - x.max()).days),
    Frequency=("Purchase Date", "count"),
    Monetary=("Total Purchase Amount", "sum")
).reset_index()

# Apply the same segment cutoffs used in the
# completed notebook to reproduce exact segments.
```

RFM Segment	Customers	Share
Needs Attention	14,492	29.18%
Loyal Customers	12,388	24.95%
Champions	8,764	17.65%
At Risk	8,644	17.41%
Recent Customers	5,373	10.82%



7. Recorded Churn Analysis

```
customer_churn = df.groupby("Customer ID")["Churn"].max()
recorded_churn_rate = customer_churn.mean() * 100
print(f"Customer-level recorded churn rate: "
      f"{recorded_churn_rate:.2f}%")
```

Observed output: Approximately 19.99% customer-level recorded churn, calculated using the maximum Churn label for each customer.

Interpretation caution: this assumes any positive transaction-level Churn label means that the customer is churned. Verify the dataset's churn definition before using the rate operationally.

8. Key Findings

• Needs Attention is the largest segment: 14,492 customers (29.18%).

• Loyal Customers account for 12,388 customers (24.95%).

• Champions account for 8,764 customers (17.65%); average recorded spending is 21,656.06.

• At Risk customers average 5.94 transactions in the summarized analysis.

• The customer-level recorded churn rate is approximately 19.99%, subject to the churn-label assumption.

9. Summary and Business Recommendations

• Re-engage Needs Attention customers with personalized reminders and relevant offers.

• Reward Champions with loyalty benefits and exclusive early access.

• Run measurable win-back campaigns for At Risk customers and track response rates.

• Encourage repeat purchases with relevant recommendations and post-purchase follow-ups.

• Monitor churn monthly and investigate additional risk factors; RFM segments alone do not explain churn.

10. Limitations

• RFM segments are rule-based groups, not a validated churn prediction model.

• The churn rate depends on customer-level aggregation of transaction labels; the churn definition should be verified.

• Returns has 47,382 missing values (18.95%); these were not imputed.

• The final month is partial, with data ending on 13 September 2023.

• The dataset currency is not specified, so total purchase amount is reported without a currency symbol.

11. Conclusion

The analysis covered 250,000 transactions from 49,661 customers and found a total recorded purchase amount of 681,346,299. RFM segmentation highlighted a large Needs Attention group alongside Loyal Customers and At Risk groups. The results support targeted engagement and retention actions, while churn conclusions should be treated cautiously until the label definition is confirmed.

Note: Code snippets are provided for documentation and readability. The executable notebook remains the source for running the complete analysis and reproducing exact segment cutoffs and all charts.